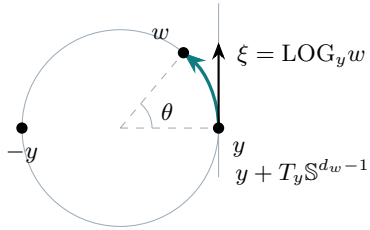
**(a) Point $\leftrightarrow$ tangent vector**

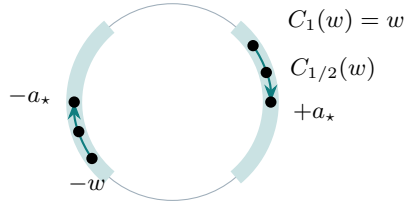



$\|\xi\| = \theta < \pi, \quad \mathrm{EXP}_y \xi = w$

The antipode $-y$ is excluded from LOG.

**(b) Contraction in two fixed caps**

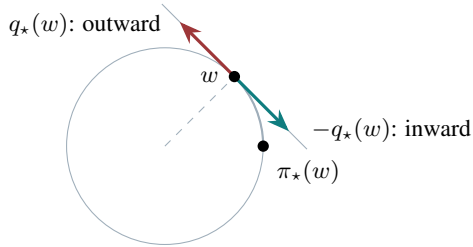
Each cap has radius $r_{\mathrm{cap}} < \pi/2$.




$C_0(w) = \pi_\star(w), \quad C_1(w) = w$

Arrows: $s$ decreases from 1 to 0.

**(c) Opposite tangent directions at the endpoint**




$q_\star(w) = \nabla_{\mathbb{S}} c_\star(w) = -\mathrm{LOG}_w \pi_\star(w)$

The inward vector is $\mathrm{LOG}_w \pi_\star(w)$.